University of Florida, Department of Mathematics

Algebra First-Year Exam, May 2025, Part 2

Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. This problem has two parts.

(a) Define the terms Euclidean domain and principal ideal.

(b) Prove that in a Euclidean domain every ideal is a principal ideal.

2. Let p be a prime and $R = \{r \in \mathbb{Q} \mid r = \frac{a}{b} \text{ for some } a, b \in \mathbb{Z} \text{ with } p \nmid b\}$. Prove that pR is the unique maximal ideal of R.

3. Using the rational canonical form, find one representative for each conjugacy class in $\mathrm{GL}(4, \mathbb{F}_2)$ whose elements have order 3 or 6. (Hint: in $\mathbb{F}_2[x]$ we have $x^6 - 1 = (x^3 - 1)^2$.)

4. Determine which of the following are irreducible in the given rings.

(a) $X^3 - X^2 + X + 2$ in $\mathbb{Q}[X]$

(b) $X^4 + X + 1$ in $\mathbb{F}_2[X]$

(c) $Y^3 + 2X^2Y + X(X+1)$ in $\mathbb{Q}[X, Y]$

5. Let F be a field, V a finite-dimensional F-vector space and $T \in \mathrm{End}_F(V)$ a linear operator.

(a) Explain why there is an F-algebra homomorphism $\psi_T : F[x] \to \mathrm{End}_F(V)$ mapping x to T.

(b) Define the minimum polynomial $m_T(x)$ of T.

(c) Let $F[T]$ be the F-subalgebra of $\mathrm{End}_F(V)$ generated by T. Prove that if $m_T(x)$ is irreducible in $F[x]$ then every nonzero element of $F[T]$ is invertible (under composition of operators).

6. Let $n \geq 1$ and let $F = \mathbb{Q}(\sqrt{1}, \sqrt{2}, \ldots, \sqrt{n})$. Show that $\sqrt[3]{2} \notin F$.